

1. Questions

Study the following data carefully and answer the questions.

A shopkeeper has five articles, A, B, C, D, and E, and the cost price of each article is different.

The ratio of the cost price of article A to that of article B is 3:5, and the ratio of the cost price of article C to that of article D is 1:2. The average of the cost prices of articles C, D, and E is equal to the cost price of article E. The difference between the cost prices of articles A and E is Rs. 150, and the difference between the cost prices of articles B and E is also Rs. 150.

The shopkeeper sold 16 units of article C, and the total profit received from selling 16 units of article C is 180% of the cost price of 1 unit of article C. The shopkeeper also sold 'N' units of article D. If the loss incurred on 1 unit of article D is Rs. 11 more than the profit received on 1 unit of article C and the total loss incurred by selling 'N' units of article D is Rs. 840, then find the value of 'N'.

- a. 20
- b. 10
- c. 15
- d. 25
- e. 5

2. Questions

Article A is marked up by 32%, and the shopkeeper received 20% profit on its selling price after giving a certain discount on its marked price. If the discount given on the marked price of article A were double the original discount, then what would have been the profit/loss percentage?

- a. 12%
- b. 5%
- c. 10%
- d. 8%
- e. Data inadequate

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3. Questions

The shopkeeper marked up article B by M%, and the marked-up amount is equal to the difference between the cost prices of articles B and C. If article B is sold at $[M - 16(2/3)]\%$ discount on its marked price, then find the profit received on article B.

- a. Rs. 37.5
- b. Rs. 20
- c. Rs. 70

d. Rs. 40

e. Rs. 7

4. Questions

The table given below shows the following data:

Article	Marked up per cent	Discount on marked price (in ₹)
A	20%	20
B	24%	120
C	30%	40
D	25%	125
E	15%	5

Find that, in which article the profit (in Rs.) is 2nd maximum?

a. C

b. A

c. D

d. B

e. E

5. Questions

From the above question, arrange the articles in descending order on the basis of profit (in Rs.) received on them:

a. $A > C > D > E > B$

b. $B > C > D > A > E$

c. $C > E > D > A > B$

d. $D > C > E > A > B$

e. $E > C > D > A > B$

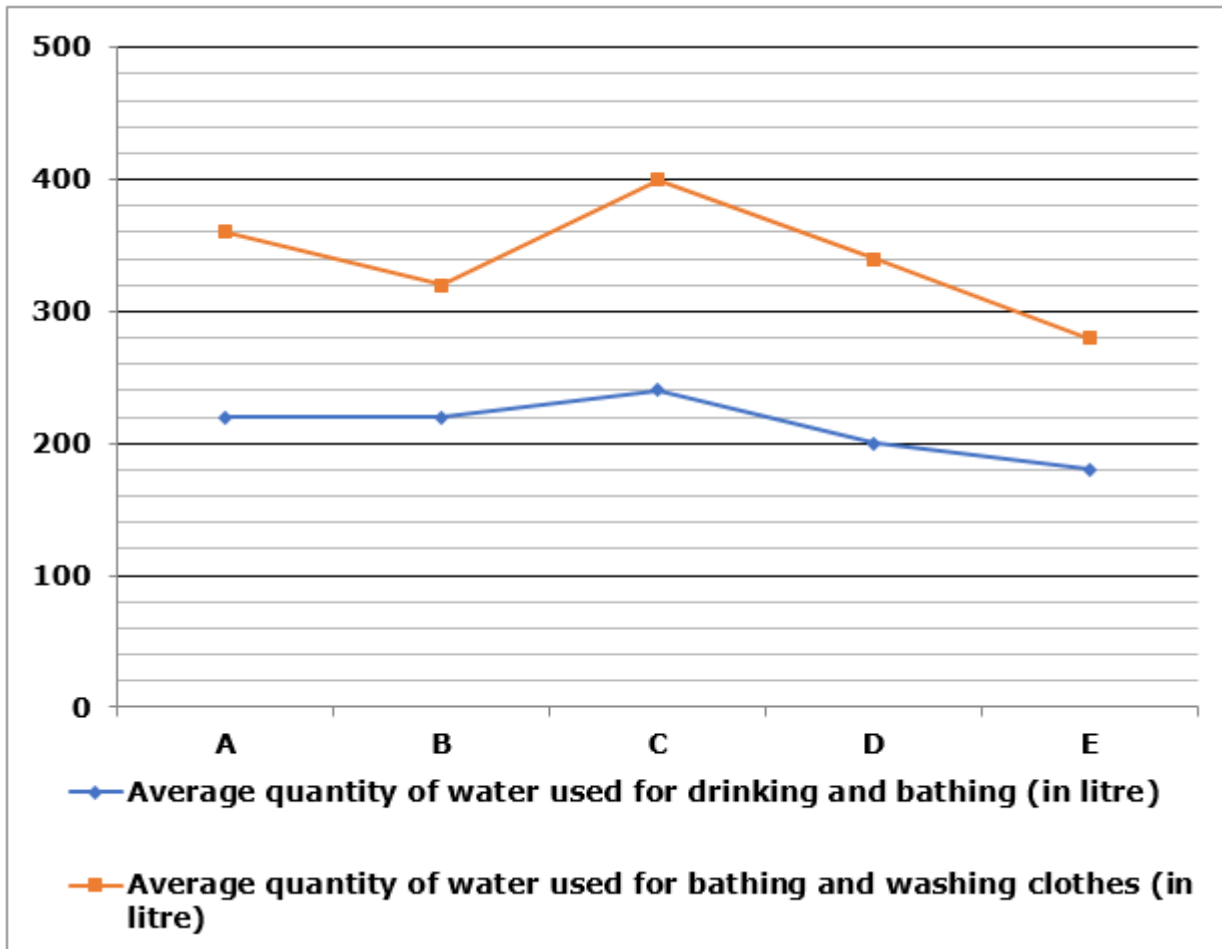
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6. Questions

Study the following data carefully and answer the questions.

Data given below is related to the total consumption (in liters) of water in a month in five families, A, B, C, D, and E. In each family, water is used for drinking, for bathing, and for washing clothes.

Line graph given below shows the following data:



If the quantities of water used for bathing in families A and C are equal and the ratio of the total consumption of water in family A to that in family C is 7:8, then find the ratio of the quantity of water used for washing clothes in family A to that in family C.

- Can't be determined
- 4: 5
- 10: 11
- 5: 6
- 20: 21

7. Questions

If the quantities of water used for drinking in families A, B, C, D, and E are respectively 160 L, 120 L, 140 L, 180 L, and 100 L, then find which of the following statements is/are not true:

P: The quantities of water used for washing clothes in families C and D are equal.

Q: The average quantity of water used for washing clothes in families B, C, and E is 320 L.

R: The quantity of water used for washing clothes in family E is $33\frac{1}{3}\%$ of that used for washing clothes in families A and D together.

- Only Q and R
- Only P

- c. Only Q
- d. Only P and R
- e. Only R

8. Questions

If the ratio of the quantity of water used for drinking to that used for washing clothes in family E is 2:7 and the cost of water is Rs. 2.5 per L, then find the total water bill of family E for that month.

- a. Rs. 1280
- b. Rs. 1600
- c. Rs. 1440
- d. Rs. 1960
- e. Rs. 1480

9. Questions

If the quantities of water used for bathing in families B and D are equal and the quantity of water used for drinking in family B is 40 L more than that used for drinking in family D, then find the total consumption of water in family B.

- a. 780 L
- b. 960 L
- c. 760 L
- d. 840 L
- e. Can't be determined

10. Questions

If in family D, the quantity of water used for washing clothes is 95% of that used for drinking and bathing together, the total consumption of water in family F is 30 L more than that in family D, and the quantity of water used for bathing in family F is $33\frac{1}{3}\%$ of the total consumption of water in family F, then find the ratio of the quantity of water used for bathing in family D to that used for bathing in family F.

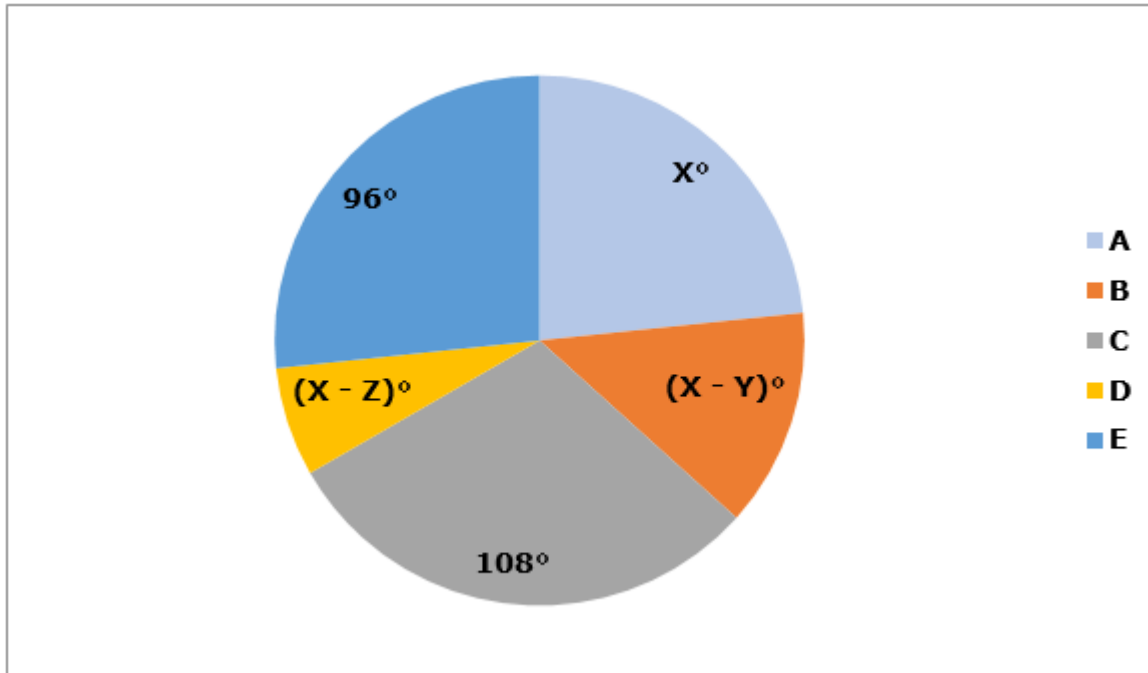
- a. 30: 29
- b. 15: 14
- c. 60: 53
- d. 5: 4
- e. 10: 9

11. Questions

Study the following data carefully and answer the questions.

A farmer has five rectangular fields (A, B, C, D and E) and the length and breadth of each field is different.

Pie chart given below shows the degree distribution of the lengths of each of the given rectangular fields.



Notes:

1: Sum of the pie chart is 750 m.

2: The length of field B is 125 m less than that of field C and also double of that of field D.

The breadth of field B is $(Z - Y)$ m, and the farmer made a 4 m wide path around field B. The farmer wants to grow grass on that path. If the cost of growing grass is Rs. $(X + Y)$ per m^2 , then find the total cost of growing grass on that path.

- a. Rs. 125280
- b. Rs. 126720
- c. Rs. 127680
- d. Rs. 126000
- e. Rs. 123000

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12. Questions

If the ratio of the breadth of field A to that of field C is 3:4 and the perimeter of field A is 4 m less than double the length of field C, then find the difference between the perimeters of both the fields.

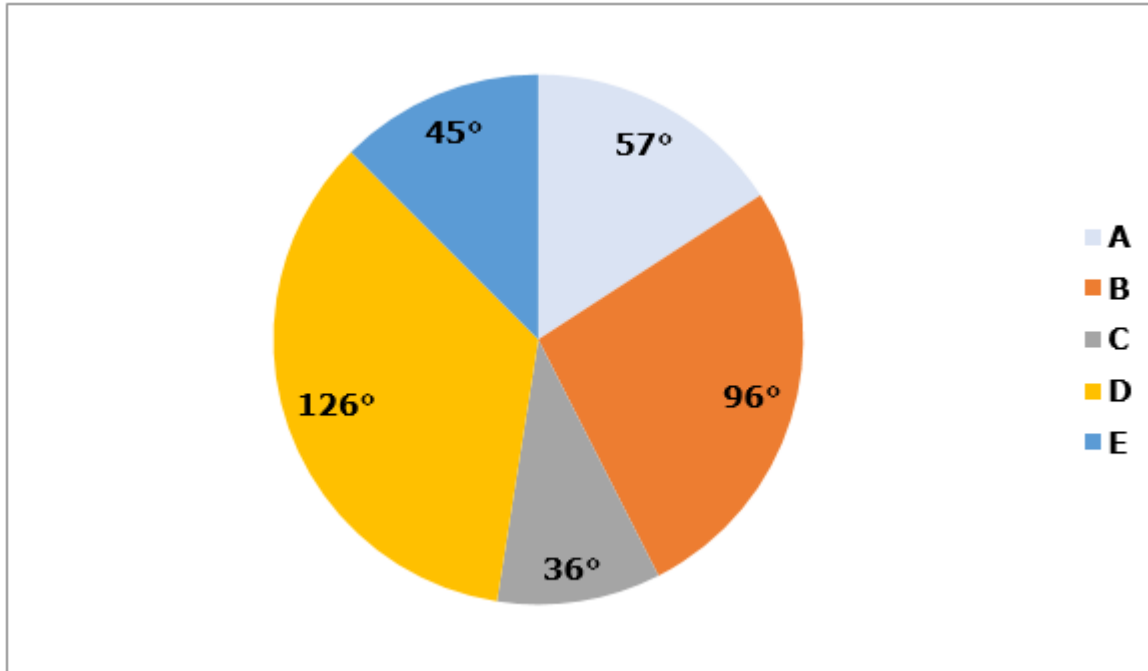
- a. $(X + Y)$ m
- b. $(X + Z - Y)$ m
- c. $(Y + Z)$ m

d. None of these

e. $(Y + 2Z) m$

13. Questions

The pie chart given below shows the degree distribution of breadths of each of the given fields.



If the difference between the breadths of fields A and E is 8 m, then find that the area of which field is maximum?

- a. Field A
- b. Field B
- c. Field C
- d. Field D
- e. Field E

14. Questions

If the breadth of field D is P m and the cost of ploughing the field is Rs. Q per m^2 , then find the total cost of ploughing field D.

Roots of $(x^2 - 93x + 1350 = 0)$ are P and Q, where $P < Q$.

- a. Rs. 64000
- b. Rs. 72500
- c. Rs. 67500
- d. Rs. 60000

e. Rs. 75000

15. Questions

If the perimeter of field E is $4(X + Y + 1)$ m and a circular field of maximum area is made inside field E, then find the remaining area of field E.

- a. Can't be determined
- b. 7014 m^2
- c. 7414 m^2
- d. 7624 m^2
- e. 7224 m^2

16. Questions

Study the following data carefully and answer the questions.

All the students from an institute are placed in five different companies, P, Q, R, S, and T, in two different designations: Junior Developer and Senior Developer.

The table given below shows the data about the number of Junior Developers and Senior Developers in the companies. Total students in the institute are 200, out of which the ratio of junior to senior developers is 3:2.

Companies	Total number of Junior Developers	Total number of Senior Developers
P	20% of total junior developers from the institute	Ratio of Senior to Junior Developers is 2: 3.
Q	4 less than that in company P.	$\frac{1}{3}$ rd of total Junior and Senior Developers in the company.
R	$\frac{3}{4}$ th of that in company P.	2^n ['n' is single digit prime number '1' more than a perfect square.]
S	----	LCM of number of Junior and Senior Developers is 24.
T	50	20% of number of Junior Developers.

What is the value(s) of 'n' are possible, and what are those value(s)?

- a. Two, 2 and 5
- b. One, 2
- c. One, 5

- d. One, Either 2 or 5
- e. Zero, Neither 2 nor 5

17. Questions

What is the ratio between the number of total developers (Junior + Senior) in the company that has the maximum number of employees and the number of total developers (Junior + Senior) in the company that has the minimum number of employees?

- a. $(n - 2^1) : (n - 3^2)$
- b. $(n - 3^1) : (n - 2^2)$
- c. $(n - 3^1) : (n - 3^2)$
- d. $(n + 2^1) : (n + 2^2)$
- e. $(n - 2^1) : (n - 2^2)$

18. Questions

M = Average of number of Senior Developers in companies R, S, and T.

L = Total number of junior developers in companies P and R together.

What is the LCM of M and L?

- a. 63
- b. 126
- c. 72
- d. 81
- e. 90

19. Questions

If some Junior and Senior developers from company P are moved to company S such that the ratio of Junior to Senior developers in companies P and S becomes 2:1 and 2:3, respectively, then find the product of the number of Junior and Senior developers that are moved.

- a. 16
- b. 32
- c. 24
- d. 20
- e. 27

20. Questions

Which of the following statements given below in the options has the highest value?

- a. The difference between the number of Junior and Senior developers in company T.
- b. Total number of Junior and Senior Developers in Company S.
- c. Sum of the number of Junior developers in companies Q and R together.
- d. Number of Senior Developers in company R.
- e. Sum of the number of Senior developers in companies P, Q, and T together.

Explanations:

1. Questions

Let the cost prices of articles A and B are Rs. $3x$ and Rs. $5x$ respectively.

Also let the cost prices of articles C and D are Rs. y and Rs. $2y$ respectively.

The cost prices of article E = $(y + 2y + \text{the cost price of article E})/3$

$3 * \text{the cost price of article E} - \text{the cost price of article E} = 3y$

$2 * \text{the cost price of article E} = 3y$

The cost price of article E = $1.5y$

Possible cases are

Since, the difference between the cost prices of articles A and E is Rs. 150 and the difference between the cost prices of articles B and E is also Rs. 150.

$$3x - 1.5y = 150 \text{ (or) } 1.5y - 3x = 150$$

$$5x - 1.5y = 150 \text{ (or) } 1.5y - 5x = 150$$

Except the equations $1.5y - 3x = 150$ and $5x - 1.5y = 150$, all must be does not satisfy

Then solve

$$1.5y - 3x = 150$$

$$5x - 1.5y = 150$$

Then solve above two equations

$$5x - 3x = 150 + 150$$

$$2x = 300$$

$$x = 150$$

and

$$5 * 150 - 1.5y = 150$$

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$$1.5y = 600$$

$$y = 400$$

Now, the cost price of article A = $3x = 3 * 150 = \text{Rs. } 450$

The cost price of article B = $5x = 5 * 150 = \text{Rs. } 750$

The cost price of article C = $y = \text{Rs. } 400$

The cost price of article D = $2y = \text{Rs. } 800$

And the cost price of article E = $1.5y = \text{Rs. } 600$

Answer: C

Since, the CP of 1 unit of article C = Rs. 400

So, the total profit received by selling 16 units of article C = 180% of 400 = Rs. 720

And the profit received on 1 unit of article C = $720/16 = \text{Rs. } 45$

Since, the loss occurred on 1 unit of article D is Rs. 11 more than the profit received on 1 unit of article C.

So, the loss occurred on 1 unit of article D = $45 + 11 = \text{Rs. } 56$

Since, the total loss occurred by selling 'N' units of article D = Rs. 840

So, the value of N = $840/56 = 15$

2. Questions

Let the cost prices of articles A and B are Rs. $3x$ and Rs. $5x$ respectively.

Also let the cost prices of articles C and D are Rs. y and Rs. $2y$ respectively.

The cost prices of article E = $(y + 2y + \text{the cost price of article E})/3$

$$3 * \text{the cost price of article E} - \text{the cost price of article E} = 3y$$

$$2 * \text{the cost price of article E} = 3y$$

$$\text{The cost price of article E} = 1.5y$$

Possible cases are

Since, the difference between the cost prices of articles A and E is Rs. 150 and the difference between the cost prices of articles B and E is also Rs. 150.

$$3x - 1.5y = 150 \text{ (or) } 1.5y - 3x = 150$$

$$5x - 1.5y = 150 \text{ (or) } 1.5y - 5x = 150$$

Except the equations $1.5y - 3x = 150$ and $5x - 1.5y = 150$, all must be does not satisfy

Then solve

$$1.5y - 3x = 150$$

$$5x - 1.5y = 150$$

Then solve above two equations

$$5x - 3x = 150 + 150$$

$$2x = 300$$

$$x = 150$$

and

$$5 * 150 - 1.5y = 150$$

$$1.5y = 600$$

$$y = 400$$

Now, the cost price of article A = $3x = 3 * 150 = \text{Rs. } 450$

The cost price of article B = $5x = 5 * 150 = \text{Rs. } 750$

The cost price of article C = $y = \text{Rs. } 400$

The cost price of article D = $2y = \text{Rs. } 800$

And the cost price of article E = $1.5y = \text{Rs. } 600$

Answer: D

Since, the CP of article A = Rs. 450

So, the MP of article A = 132% of 450 = Rs. 594

Let the SP of the article A = Rs. a

Since, the profit is 20% of the SP.

So,

$$(a - 450) = 20\% \text{ of } 450$$

$$a - 450 = 90$$

$$a = 540$$

Since, the SP of article A = Rs. 540

So, the discount given on the MP of article A = $594 - 540 = \text{Rs. } 54$

Since, the new discount given on the MP of article A = $2 * 54 = \text{Rs. } 108$

So, the new SP of article A = $594 - 108 = \text{Rs. } 486$

And the profit per cent = $[(486 - 450)/450] * 100 = 8\%$

3. Questions

Let the cost prices of articles A and B are Rs. $3x$ and Rs. $5x$ respectively.

Also let the cost prices of articles C and D are Rs. y and Rs. $2y$ respectively.

The cost prices of article E = $(y + 2y + \text{the cost price of article E})/3$

3 * the cost price of article E - the cost price of article E = 3y

2 * the cost price of article E = 3y

The cost price of article E = 1.5y

Possible cases are

Since, the difference between the cost prices of articles A and E is Rs. 150 and the difference between the cost prices of articles B and E is also Rs. 150.

$$3x - 1.5y = 150 \text{ (or) } 1.5y - 3x = 150$$

$$5x - 1.5y = 150 \text{ (or) } 1.5y - 5x = 150$$

Except the equations $1.5y - 3x = 150$ and $5x - 1.5y = 150$, all must be does not satisfy

Then solve

$$1.5y - 3x = 150$$

$$5x - 1.5y = 150$$

Then solve above two equations

$$5x - 3x = 150 + 150$$

$$2x = 300$$

$$x = 150$$

and

$$5 * 150 - 1.5y = 150$$

$$1.5y = 600$$

$$y = 400$$

Now, the cost price of article A = $3x = 3 * 150 = \text{Rs. } 450$

The cost price of article B = $5x = 5 * 150 = \text{Rs. } 750$

The cost price of article C = $y = \text{Rs. } 400$

The cost price of article D = $2y = \text{Rs. } 800$

And the cost price of article E = $1.5y = \text{Rs. } 600$

Answer: B

Since, the CP of article B = Rs. 750

And the difference between the cost prices of articles B and C = $750 - 400 = \text{Rs. } 350$

So, the MP of article B = $750 + 350 = \text{Rs. } 1100$

And, $M = (350/750) * 100 = 46(2/3)\%$

Since, the discount on the MP of article B = $[M - 16(2/3)]\% = [46(2/3) - 16(2/3)]\% = 30\%$

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So, the SP of article B = 70% of 1100 = Rs. 770

And the profit received on article B = 770 – 750 = Rs. 20

4. Questions

Let the cost prices of articles A and B are Rs. 3x and Rs. 5x respectively.

Also let the cost prices of articles C and D are Rs. y and Rs. 2y respectively.

The cost prices of article E = (y + 2y + the cost price of article E)/3

3 * the cost price of article E - the cost price of article E = 3y

2 * the cost price of article E = 3y

The cost price of article E = 1.5y

Possible cases are

Since, the difference between the cost prices of articles A and E is Rs. 150 and the difference between the cost prices of articles B and E is also Rs. 150.

$3x - 1.5y = 150$ (or) $1.5y - 3x = 150$

$5x - 1.5y = 150$ (or) $1.5y - 5x = 150$

Except the equations $1.5y - 3x = 150$ and $5x - 1.5y = 150$, all must be does not satisfy

Then solve

$1.5y - 3x = 150$

$5x - 1.5y = 150$

Then solve above two equations

$5x - 3x = 150 + 150$

$2x = 300$

$x = 150$

and

$5 * 150 - 1.5y = 150$

$1.5y = 600$

$y = 400$

Now, the cost price of article A = $3x = 3 * 150 =$ Rs. 450

The cost price of article B = $5x = 5 * 150 =$ Rs. 750

The cost price of article C = $y =$ Rs. 400

The cost price of article D = $2y =$ Rs. 800

And the cost price of article E = $1.5y =$ Rs. 600

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Answer: A

Article	Cost price (in ₹)	Marked price (in ₹)	Selling price (in ₹)	Profit received (in ₹)
A	450	120% of 450 = 540	540 – 20 = 520	520 – 450 = 70
B	750	124% of 750 = 930	930 – 120 = 810	810 – 750 = 60
C	400	130% of 400 = 520	520 – 40 = 480	480 – 400 = 80
D	800	125% of 800 = 1000	1000 – 125 = 875	875 – 800 = 75
E	600	115% of 600 = 690	690 – 5 = 685	685 – 600 = 85

So, on article C, the received profit is 2nd maximum.

5. Questions

Let the cost prices of articles A and B are Rs. 3x and Rs. 5x respectively.

Also let the cost prices of articles C and D are Rs. y and Rs. 2y respectively.

The cost prices of article E = (y + 2y + the cost price of article E)/3

3 * the cost price of article E - the cost price of article E = 3y

2 * the cost price of article E = 3y

The cost price of article E = 1.5y

Possible cases are

Since, the difference between the cost prices of articles A and E is Rs. 150 and the difference between the cost prices of articles B and E is also Rs. 150.

$3x - 1.5y = 150$ (or) $1.5y - 3x = 150$

$5x - 1.5y = 150$ (or) $1.5y - 5x = 150$

Except the equations $1.5y - 3x = 150$ and $5x - 1.5y = 150$, all must be does not satisfy

Then solve

$1.5y - 3x = 150$

$5x - 1.5y = 150$

Then solve above two equations

$5x - 3x = 150 + 150$

$2x = 300$

$x = 150$

and

$5 * 150 - 1.5y = 150$

$1.5y = 600$

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$$y = 400$$

Now, the cost price of article A = $3x = 3 * 150 = \text{Rs. } 450$

The cost price of article B = $5x = 5 * 150 = \text{Rs. } 750$

The cost price of article C = $y = \text{Rs. } 400$

The cost price of article D = $2y = \text{Rs. } 800$

And the cost price of article E = $1.5y = \text{Rs. } 600$

Answer: E

The profit received on article A = Rs. 70

The profit received on article B = Rs. 60

The profit received on article C = Rs. 80

The profit received on article D = Rs. 75

The profit received on article E = Rs. 85

So, the required order: $E > C > D > A > B$

6. Questions

Family	The total quantity of water used for drinking and bathing (in L)	The total quantity of water used for bathing and washing clothes (in L)
A	$2 * 220 = 440$	$2 * 360 = 720$
B	$2 * 220 = 440$	$2 * 320 = 640$
C	$2 * 240 = 480$	$2 * 400 = 800$
D	$2 * 200 = 400$	$2 * 340 = 680$
E	$2 * 180 = 360$	$2 * 280 = 560$

Answer: D

Let the quantity of water used for bathing in family A = the quantity of water used for bathing in family C = 'x' L

So, the total consumption of water in family A = $[(440 - x) + (720 - x) + x] = (1160 - x)$ L

And the total consumption of water in family C = $[(480 - x) + (800 - x) + x] = (1280 - x)$ L

So,

$$(1160 - x)/(1280 - x) = 7/8$$

$$9280 - 8x = 8960 - 7x$$

$$x = 320$$

Since, the quantity of water used for washing clothes in family A = $720 - 320 = 400$ L

And the quantity of water used for washing clothes in family C = $800 - 320 = 480$ L

So, the required ratio = $400:480 = 5:6$

7. Questions

Family	The total quantity of water used for drinking and bathing (in L)	The total quantity of water used for bathing and washing clothes (in L)
A	$2 * 220 = 440$	$2 * 360 = 720$
B	$2 * 220 = 440$	$2 * 320 = 640$
C	$2 * 240 = 480$	$2 * 400 = 800$
D	$2 * 200 = 400$	$2 * 340 = 680$
E	$2 * 180 = 360$	$2 * 280 = 560$

Answer: C

Family	The quantity of water used for drinking (in L)	The quantity of water used for bathing (in L)	The quantity of water used for washing clothes (in L)
A	160	$440 - 160 = 280$	$720 - 280 = 440$
B	120	$440 - 120 = 320$	$640 - 320 = 320$
C	140	$480 - 140 = 340$	$800 - 340 = 460$
D	180	$400 - 180 = 220$	$680 - 220 = 460$
E	100	$360 - 100 = 260$	$560 - 260 = 300$

From P:

Since, the quantities of water used for washing clothes in families C and D are equal, which is 460 L.

So, P is true.

From Q:

Average quantity of water used for washing clothes in families B, C and E = $(320 + 460 + 300)/3 = 360$ L

So, Q is not true.

From R:

Quantity of water used for washing clothes in families A and D together = $440 + 460 = 900$ L

Quantity of water used for washing clothes in family E = 300 L

Since, the required percentage = $(300/900) * 100 = 33(1/3)\%$

So, R is true.

Hence, only Q is not true.

8. Questions

Family	The total quantity of water used for drinking and bathing (in L)	The total quantity of water used for bathing and washing clothes (in L)
A	$2 * 220 = 440$	$2 * 360 = 720$
B	$2 * 220 = 440$	$2 * 320 = 640$
C	$2 * 240 = 480$	$2 * 400 = 800$
D	$2 * 200 = 400$	$2 * 340 = 680$
E	$2 * 180 = 360$	$2 * 280 = 560$

Answer: B

Let the quantities of water used for drinking and washing clothes in family E are '2m' L and '7m' L respectively.

Also let the quantity of water used for bathing in family E = 'n' L

So, $2m + n = 360$ -----(1)

And, $7m + n = 560$ -----(2)

From equations (1) and (2):

$$360 - 2m = 560 - 7m$$

$$m = 40$$

From equation (1):

$$2 * 40 + n = 360$$

$$n = 280$$

Since, the total consumption of water in family E = $(2 * 40) + 280 + (7 * 40) = 640$ L

So, the total water bill of family E for that month = $640 * 2.5 = \text{Rs. } 1600$

9. Questions

Family	The total quantity of water used for drinking and bathing (in L)	The total quantity of water used for bathing and washing clothes (in L)
A	$2 * 220 = 440$	$2 * 360 = 720$
B	$2 * 220 = 440$	$2 * 320 = 640$
C	$2 * 240 = 480$	$2 * 400 = 800$
D	$2 * 200 = 400$	$2 * 340 = 680$
E	$2 * 180 = 360$	$2 * 280 = 560$

Answer: E

Let the quantity of water used for bathing in family B = the quantity of water used for bathing in family D = 'y' L

So, the quantity of water used for drinking in family B = $(440 - y)$ L

And the quantity of water used for drinking in family D = $(400 - y)$ L

Since, the quantity of water used for drinking in family B is 40 L more than that used for drinking in family D

So,

$$(440 - y) - (400 - y) = 40$$

$$440 - y - 400 + y = 40$$

Since, we can't calculate the value of 'y'.

So, the required answer can't be determined.

10. Questions

Family	The total quantity of water used for drinking and bathing (in L)	The total quantity of water used for bathing and washing clothes (in L)
A	$2 * 220 = 440$	$2 * 360 = 720$
B	$2 * 220 = 440$	$2 * 320 = 640$
C	$2 * 240 = 480$	$2 * 400 = 800$
D	$2 * 200 = 400$	$2 * 340 = 680$
E	$2 * 180 = 360$	$2 * 280 = 560$

Answer: E

Since, the quantity of water used for drinking and bathing together in family D = 400 L

So, the quantity of water used for washing clothes in family D = 95% of 400 = 380 L

The quantity of water used for bathing in family D = $680 - 380 = 300$ L

The quantity of water used for drinking in family D = $400 - 300 = 100$ L

And the total consumption of water in family D = $100 + 300 + 380 = 780$ L

Since, the total consumption of water in family F = $780 + 30 = 810$ L

And the quantity of water used for bathing in family F = $33\frac{1}{3}\%$ of 810 = 270 L

So, the required ratio = $300 : 270 = 10 : 9$

11. Questions

Since, the sum of the pie chart = 750 m

So, the length of field C = $750 * (108/360) = 225$ m

Since, the length of field B is 125 m less than that of field C.

So, the length of field B = $225 - 125 = 100$ m

And, $(X - Y) = 360 * (100/750)$

$$(X - Y) = 48 \text{ -----(1)}$$

Since, the length of field B is double of that of field D.

$$\text{So, the length of field D} = 100/2 = 50 \text{ m}$$

$$\text{And, } (X - Z) = 360 * (50/750)$$

$$(X - Z) = 24 \text{ -----(2)}$$

From the pie chart:

$$X + (X - Y) + 108 + (X - Z) + 96 = 360$$

$$X + (X - Y) + (X - Z) = 156 \text{ -----(3)}$$

From equations (1), (2) and (3):

$$X + 48 + 24 = 156$$

$$X = 84$$

From equation (1):

$$84 - Y = 48$$

$$Y = 36$$

From equation (2):

$$84 - Z = 24$$

$$Z = 60$$

$$\text{Now, the length of field A} = 750 * (84/360) = 175 \text{ m}$$

$$\text{The length of field B} = 750 * (48/360) = 100 \text{ m}$$

$$\text{The length of field C} = 750 * (108/360) = 225 \text{ m}$$

$$\text{The length of field D} = 750 * (24/360) = 50 \text{ m}$$

$$\text{And the length of field E} = 750 * (96/360) = 200 \text{ m}$$

Answer: B

$$\text{The length of field B} = 100 \text{ m}$$

$$\text{The breadth of field B} = (Z - Y) = (60 - 36) = 24 \text{ m}$$

Since, the farmer made a 4 m wide path around field B.

$$\text{So, the area of field B without path} = 100 * 24 = 2400 \text{ m}^2$$

$$\text{The area of field B with the path} = (100 + 8) * (24 + 8) = 3456 \text{ m}^2$$

$$\text{And the area of the path} = 3456 - 2400 = 1056 \text{ m}^2$$

$$\text{Since, the cost of growing grass} = (X + Y) = (84 + 36) = \text{Rs. } 120 \text{ per m}^2$$

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So, the total cost of growing grass on the path = $1056 * 120 = \text{Rs. } 126720$

12. Questions

Since, the sum of the pie chart = 750 m

So, the length of field C = $750 * (108/360) = 225 \text{ m}$

Since, the length of field B is 125 m less than that of field C.

So, the length of field B = $225 - 125 = 100 \text{ m}$

And, $(X - Y) = 360 * (100/750)$

$(X - Y) = 48$ -----(1)

Since, the length of field B is double of that of field D.

So, the length of field D = $100/2 = 50 \text{ m}$

And, $(X - Z) = 360 * (50/750)$

$(X - Z) = 24$ -----(2)

From the pie chart:

$X + (X - Y) + 108 + (X - Z) + 96 = 360$

$X + (X - Y) + (X - Z) = 156$ -----(3)

From equations (1), (2) and (3):

$X + 48 + 24 = 156$

$X = 84$

From equation (1):

$84 - Y = 48$

$Y = 36$

From equation (2):

$84 - Z = 24$

$Z = 60$

Now, the length of field A = $750 * (84/360) = 175 \text{ m}$

The length of field B = $750 * (48/360) = 100 \text{ m}$

The length of field C = $750 * (108/360) = 225 \text{ m}$

The length of field D = $750 * (24/360) = 50 \text{ m}$

And the length of field E = $750 * (96/360) = 200 \text{ m}$

Answer: D

Let the breadths of fields A and C be '3b' m and '4b' m respectively.

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Since, the length of field A = 175 m

So, the perimeter of field A = $2(175 + 3b) = (350 + 6b)$ m

Since, the length of field B = 225 m

So, the perimeter of field B = $2(225 + 4b) = (450 + 8b)$ m

Since, the perimeter of field A is 4 m less than the double of the length of field C.

So,

$$350 + 6b = 2 * 225 - 4$$

$$350 + 6b = 446$$

$$b = 16$$

Since, the perimeter of field A = $(350 + 6 * 16) = 446$ m

And the perimeter of field C = $(450 + 8 * 16) = 578$ m

So, the required difference = $578 - 446 = 132$ m

Since, $(X + Y) = 84 + 36 = 120$

$(X + Z - Y) = (84 + 60 - 36) = 108$

$(Y + Z) = (36 + 60) = 96$

And, $(Y + 2Z) = (36 + 120) = 156$

So, option (d) is correct answer.

13. Questions

Since, the sum of the pie chart = 750 m

So, the length of field C = $750 * (108/360) = 225$ m

Since, the length of field B is 125 m less than that of field C.

So, the length of field B = $225 - 125 = 100$ m

And, $(X - Y) = 360 * (100/750)$

$$(X - Y) = 48 \text{ -----(1)}$$

Since, the length of field B is double of that of field D.

So, the length of field D = $100/2 = 50$ m

And, $(X - Z) = 360 * (50/750)$

$$(X - Z) = 24 \text{ -----(2)}$$

From the pie chart:

$$X + (X - Y) + 108 + (X - Z) + 96 = 360$$

$$X + (X - Y) + (X - Z) = 156 \text{ -----(3)}$$

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From equations (1), (2) and (3):

$$X + 48 + 24 = 156$$

$$X = 84$$

From equation (1):

$$84 - Y = 48$$

$$Y = 36$$

From equation (2):

$$84 - Z = 24$$

$$Z = 60$$

Now, the length of field A = $750 * (84/360) = 175$ m

The length of field B = $750 * (48/360) = 100$ m

The length of field C = $750 * (108/360) = 225$ m

The length of field D = $750 * (24/360) = 50$ m

And the length of field E = $750 * (96/360) = 200$ m

Answer: A

Since, the difference between the breadths of fields A and E = 8 m

So, the breadth of field A = $8 * [57/(57 - 45)] = 38$ m

The breadth of field B = $38 * (96/57) = 64$ m

The breadth of field C = $38 * (36/57) = 24$ m

The breadth of field D = $38 * (126/57) = 84$ m

And the breadth of field E = $38 * (45/57) = 30$ m

Field	Length (in m)	Breadth (in m)	Area (in m ²)
A	175	38	$175 * 38 = 6650$
B	100	64	$100 * 64 = 6400$
C	225	24	$225 * 24 = 5400$
D	50	84	$50 * 84 = 4200$
E	200	30	$200 * 30 = 6000$

So, the area of field A is maximum.

14. Questions

Since, the sum of the pie chart = 750 m

So, the length of field C = $750 * (108/360) = 225$ m

Since, the length of field B is 125 m less than that of field C.

So, the length of field B = $225 - 125 = 100$ m

And, $(X - Y) = 360 * (100/750)$

$(X - Y) = 48$ -----(1)

Since, the length of field B is double of that of field D.

So, the length of field D = $100/2 = 50$ m

And, $(X - Z) = 360 * (50/750)$

$(X - Z) = 24$ -----(2)

From the pie chart:

$X + (X - Y) + 108 + (X - Z) + 96 = 360$

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From equations (1), (2) and (3):

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$X = 84$

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$Z = 60$

Now, the length of field A = $750 * (84/360) = 175$ m

The length of field B = $750 * (48/360) = 100$ m

The length of field C = $750 * (108/360) = 225$ m

The length of field D = $750 * (24/360) = 50$ m

And the length of field E = $750 * (96/360) = 200$ m

Answer: C

From the equation:

$$x^2 - 93x + 1350 = 0$$

$$x^2 - 75x - 18x + 1350 = 0$$

$$x(x - 75) - 18(x - 75) = 0$$

$$x = 18, 75$$

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So, $P = 18$ and $Q = 75$

Since, the breadth of field D = $P \text{ m} = 18 \text{ m}$

And the length of field D = 50 m

So, the area of field D = $50 * 18 = 900 \text{ m}^2$

Since, the cost of ploughing the field = Rs. $Q \text{ per m}^2 = \text{Rs. } 75 \text{ per m}^2$

So, the total cost of ploughing field D = $900 * 75 = \text{Rs. } 67500$

15. Questions

Since, the sum of the pie chart = 750 m

So, the length of field C = $750 * (108/360) = 225 \text{ m}$

Since, the length of field B is 125 m less than that of field C.

So, the length of field B = $225 - 125 = 100 \text{ m}$

And, $(X - Y) = 360 * (100/750)$

$(X - Y) = 48$ -----(1)

Since, the length of field B is double of that of field D.

So, the length of field D = $100/2 = 50 \text{ m}$

And, $(X - Z) = 360 * (50/750)$

$(X - Z) = 24$ -----(2)

From the pie chart:

$X + (X - Y) + 108 + (X - Z) + 96 = 360$

$X + (X - Y) + (X - Z) = 156$ -----(3)

From equations (1), (2) and (3):

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$X = 84$

From equation (1):

$84 - Y = 48$

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Now, the length of field A = $750 * (84/360) = 175 \text{ m}$

The length of field B = $750 * (48/360) = 100 \text{ m}$

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The length of field C = $750 * (108/360) = 225$ m

The length of field D = $750 * (24/360) = 50$ m

And the length of field E = $750 * (96/360) = 200$ m

Answer: B

Since, the perimeter of field E = $4(X + Y + 1) = 4(84 + 36 + 1) = 484$ m

And the length of field E = 200 m

So, the breadth of field E = $(484 - 2 * 200)/2 = 42$ m

Since, a circular field of maximum area is made inside field E.

So, the diameter of the circular field will be equal to the breadth of field E.

Since, the diameter of the circular field = 42 m

So, the radius of the circular field = $42/2 = 21$ m

And the area of the circular field = $(22/7) * 21^2 = 1386$ m²

Since, the area of field E = $200 * 42 = 8400$ m²

So, remaining area of field E = $8400 - 1386 = 7014$ m²

16. Questions

Total students in the institute = 200

Total number of Junior Developers in the institute = $200 * 3/5 = 120$

Total number of Senior Developers in the institute = $200 * 2/5 = 80$

Companies	Total number of Junior Developers	Total number of Senior Developers
P	20% of 120 = 24	$24 * 2/3 = 16$
Q	$24 - 4 = 20$	$20 * 1/(3 - 1) = 20 * 1/2 = 10$
R	$3/4 * 24 = 18$	---
S	---	---
T	50	$20\% \text{ of } 50 = 10$

We calculate the number of Junior Developers and difference in the companies in which it is easy to calculate. Now, we need to calculate for other companies.

Number of Junior Developers in company S = $120 - 24 - 20 - 18 - 50 = 8$

Since, LCM of Junior and Senior Developers in company S is 24.

So, possible value(s) of number of Senior Developers in company S can be 6, 12 and 24.

'n' is single digit prime number '1' more than a perfect square.

Single digit prime numbers = 2, 3, 5, and 7 out of which 2 and 5 satisfy the above given condition.

So, possible value(s) of number of Senior Developers in company R can be $2^2 (=4)$ and $2^5 (= 32)$.

Possible pairs of number of Senior Developers in companies R and S = (4, 6), (4, 12), (4, 24), (32, 6), (32, 12), and (32, 24)

Sum of number of Senior Developers in companies R and S = $80 - 16 - 10 - 10 = 44$

So, correct pair of number of Senior Developers in companies R and S = (12, 32)

Number of Senior Developers in company R = 32

Number of Senior Developers in company S = 12

Companies	Total number of Junior Developers	Total number of Senior Developers
P	24	16
Q	20	10
R	18	32
S	8	12
T	50	10

Answer: C

Number of Senior Developers in company R = $2^n = 32$

$$2^n = 2^5$$

$$n = 5$$

Hence, only one value of 'n' is possible which is 5.

17. Questions

Total students in the institute = 200

Total number of Junior Developers in the institute = $200 * 3/5 = 120$

Total number of Senior Developers in the institute = $200 * 2/5 = 80$

Companies	Total number of Junior Developers	Total number of Senior Developers
P	$20\% \text{ of } 120 = 24$	$24 * 2/3 = 16$
Q	$24 - 4 = 20$	$20 * 1/(3 - 1) = 20 * 1/2 = 10$
R	$3/4 * 24 = 18$	----
S	----	----
T	50	$20\% \text{ of } 50 = 10$

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We calculate the number of Junior Developers and difference in the companies in which it is easy to calculate. Now, we need to calculate for other companies.

Number of Junior Developers in company S = $120 - 24 - 20 - 18 - 50 = 8$

Since, LCM of Junior and Senior Developers in company S is 24.

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'n' is single digit prime number '1' more than a perfect square.

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So, possible value(s) of number of Senior Developers in company R can be $2^2 (=4)$ and $2^5 (=32)$.

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Sum of number of Senior Developers in companies R and S = $80 - 16 - 10 - 10 = 44$

So, correct pair of number of Senior Developers in companies R and S = (12, 32)

Number of Senior Developers in company R = 32

Number of Senior Developers in company S = 12

Companies	Total number of Junior Developers	Total number of Senior Developers
P	24	16
Q	20	10
R	18	32
S	8	12
T	50	10

Answer: E

Total employees in company P = $24 + 16 = 40$

Total employees in company Q = $20 + 10 = 30$

Total employees in company R = $18 + 32 = 50$

Total employees in company S = $8 + 12 = 20$

Total employees in company T = $50 + 10 = 60$

Since companies S and T has minimum and maximum number of employees in it.

Required ratio = 60: 20

= 3: 1

= $(n - 2): (n - 4)$

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$$= (n - 2^1): (n - 2^2)$$

18. Questions

Total students in the institute = 200

Total number of Junior Developers in the institute = $200 * \frac{3}{5} = 120$

Total number of Senior Developers in the institute = $200 * \frac{2}{5} = 80$

Companies	Total number of Junior Developers	Total number of Senior Developers
P	20% of 120 = 24	$24 * \frac{2}{3} = 16$
Q	$24 - 4 = 20$	$20 * \frac{1}{(3 - 1)} = 20 * \frac{1}{2} = 10$
R	$\frac{3}{4} * 24 = 18$	----
S	----	----
T	50	20% of 50 = 10

We calculate the number of Junior Developers and difference in the companies in which it is easy to calculate. Now, we need to calculate for other companies.

Number of Junior Developers in company S = $120 - 24 - 20 - 18 - 50 = 8$

Since, LCM of Junior and Senior Developers in company S is 24.

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'n' is single digit prime number '1' more than a perfect square.

Single digit prime numbers = 2, 3, 5, and 7 out of which 2 and 5 satisfy the above given condition.

So, possible value(s) of number of Senior Developers in company R can be $2^2 (=4)$ and $2^5 (= 32)$.

Possible pairs of number of Senior Developers in companies R and S = (4, 6), (4, 12), (4, 24), (32, 6), (32, 12), and (32, 24)

Sum of number of Senior Developers in companies R and S = $80 - 16 - 10 - 10 = 44$

So, correct pair of number of Senior Developers in companies R and S = (12, 32)

Number of Senior Developers in company R = 32

Number of Senior Developers in company S = 12

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Companies	Total number of Junior Developers	Total number of Senior Developers
P	24	16
Q	20	10
R	18	32
S	8	12
T	50	10

Answer: B

M = Average of number of Senior Developers in companies R, S, and T = $(32 + 12 + 10)/3$

= $54/3 = 18$

L = Total number of Junior Developers in companies P and R together = $24 + 18$

= 42

LCM of M and N = LCM of 18 and 42

$18 = 2 * 3^2$

$42 = 2 * 3 * 7$

LCM of 18 and 42 = $2 * 3^2 * 7$

= 126

19. Questions

Total students in the institute = 200

Total number of Junior Developers in the institute = $200 * 3/5 = 120$

Total number of Senior Developers in the institute = $200 * 2/5 = 80$

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P	20% of 120 = 24	$24 * 2/3 = 16$
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R	$3/4 * 24 = 18$	----
S	----	----
T	50	20% of 50 = 10

We calculate the number of Junior Developers and difference in the companies in which it is easy to calculate. Now, we need to calculate for other companies.

Number of Junior Developers in company S = $120 - 24 - 20 - 18 - 50 = 8$

Since, LCM of Junior and Senior Developers in company S is 24.

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Number of Senior Developers in company R = 32

Number of Senior Developers in company S = 12

Companies	Total number of Junior Developers	Total number of Senior Developers
P	24	16
Q	20	10
R	18	32
S	8	12
T	50	10

Answer: C

Let number of Junior and Senior Developers those are moved from company P to company S are 'a' and 'b' respectively.

According to the question:

$$(24 - a) : (16 - b) = 2 : 1$$

$$24 - a = 32 - 2b$$

$$2b - a = 8 \dots\dots\dots (1)$$

$$(8 + a) : (12 + b) = 2 : 3$$

$$24 + 3a = 24 + 2b$$

$$3a = 2b \dots\dots\dots (2)$$

From (1) and (2):

$$3a - a = 8$$

$$a = 4 \text{ and } b = 6$$

$$\text{Required product} = a * b = 4 * 6$$

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= 24

20. Questions

Total students in the institute = 200

Total number of Junior Developers in the institute = $200 \times \frac{3}{5} = 120$

Total number of Senior Developers in the institute = $200 \times \frac{2}{5} = 80$

Companies	Total number of Junior Developers	Total number of Senior Developers
P	$20\% \text{ of } 120 = 24$	$24 \times \frac{2}{3} = 16$
Q	$24 - 4 = 20$	$20 \times \frac{1}{(3-1)} = 20 \times \frac{1}{2} = 10$
R	$\frac{3}{4} \times 24 = 18$	----
S	----	----
T	50	$20\% \text{ of } 50 = 10$

We calculate the number of Junior Developers and difference in the companies in which it is easy to calculate. Now, we need to calculate for other companies.

Number of Junior Developers in company S = $120 - 24 - 20 - 18 - 50 = 8$

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So, possible value(s) of number of Senior Developers in company R can be $2^2 (=4)$ and $2^5 (= 32)$.

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Sum of number of Senior Developers in companies R and S = $80 - 16 - 10 - 10 = 44$

So, correct pair of number of Senior Developers in companies R and S = (12, 32)

Number of Senior Developers in company R = 32

Number of Senior Developers in company S = 12

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Companies	Total number of Junior Developers	Total number of Senior Developers
P	24	16
Q	20	10
R	18	32
S	8	12
T	50	10

Answer: A

Option (a): Difference between number of Junior and Senior Developers in company T = $50 - 10 = 40$

Option (b): Total number of Junior and Senior Developers in Company S = $8 + 12 = 20$

Option (c): Sum of number of Junior Developers in companies Q and R together = $20 + 18 = 38$

Option (d): Number of Senior Developers in company R = 32

Option (e): Sum of number of Senior Developers in companies P, Q, and T together = $16 + 10 + 10 = 36$

Hence, the statement given in option (a) has the highest value.

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